PROJECT “OUROBOROS”

OR SNAKE THE GAME

STUDENT PRACTICE

OUR TEAM

Artur Bodrov

- Teamlead
- PC protocol developer
- UI developer backup

Rostyslav Kazuta

- UI developer
- PC protocol developer backup

Vasyl Polishchuck

- STM driver developer
- Backend developer backup

Vaskiv Valentyn

- Backend developer
- STM driver developer backup

Organization of work

Before(3 week)

Backlog | Priority board | Team items | Roadmap | My items | + New view

Filter by keyword or by field

Backlog 3 / 5 Estimate: 0
This item hasn't been started

- student_work_snake_game #25
  Design and implement error handling.
- student_work_snake_game #26
  Design and implement log approach
- student_work_snake_game #27
  Move settings to controller

+ Add item

Ready 0 Estimate: 0
This is ready to be picked up

+ Add item

In progress 9 / 3 Estimate: 0
This is actively being worked on

- Bug fix UI painting wall
- student_work_snake_game #30
  Bug fix create backend level
- student_work_snake_game #32
  Bug fix default speed value
- student_work_snake_game #31
  Creating a presentation for a project
- student_work_snake_game #28
  Add configuration to for scanning COM port

+ Add item

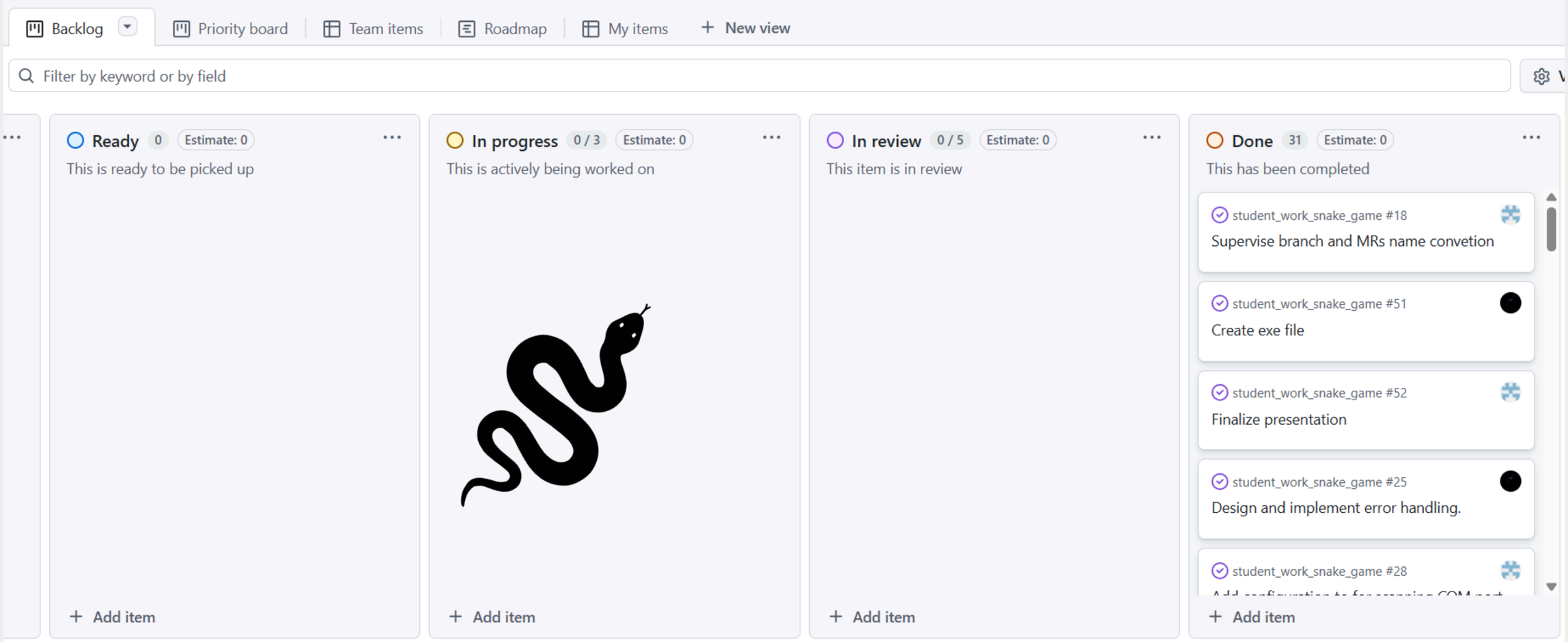
In review 4 / 5 Estimate: 0
This item is in review

- student_work_snake_game #16
  Fix Issue with backend of game
- student_work_snake_game #23
  Update UART protcol part 1
- student_work_snake_game #20
  Design and implement obstacle changing from stm side
- student_work_snake_game #19
  Design and implement speed changing from STM side

+ Add item

Organization of work

After



Statistics

Backlog | Priority board | Team items | Roadmap | My items | + New view

Assignees

Alko2307	10
RareT128	10
Rozstyslav	10
valentynvaskiv	10

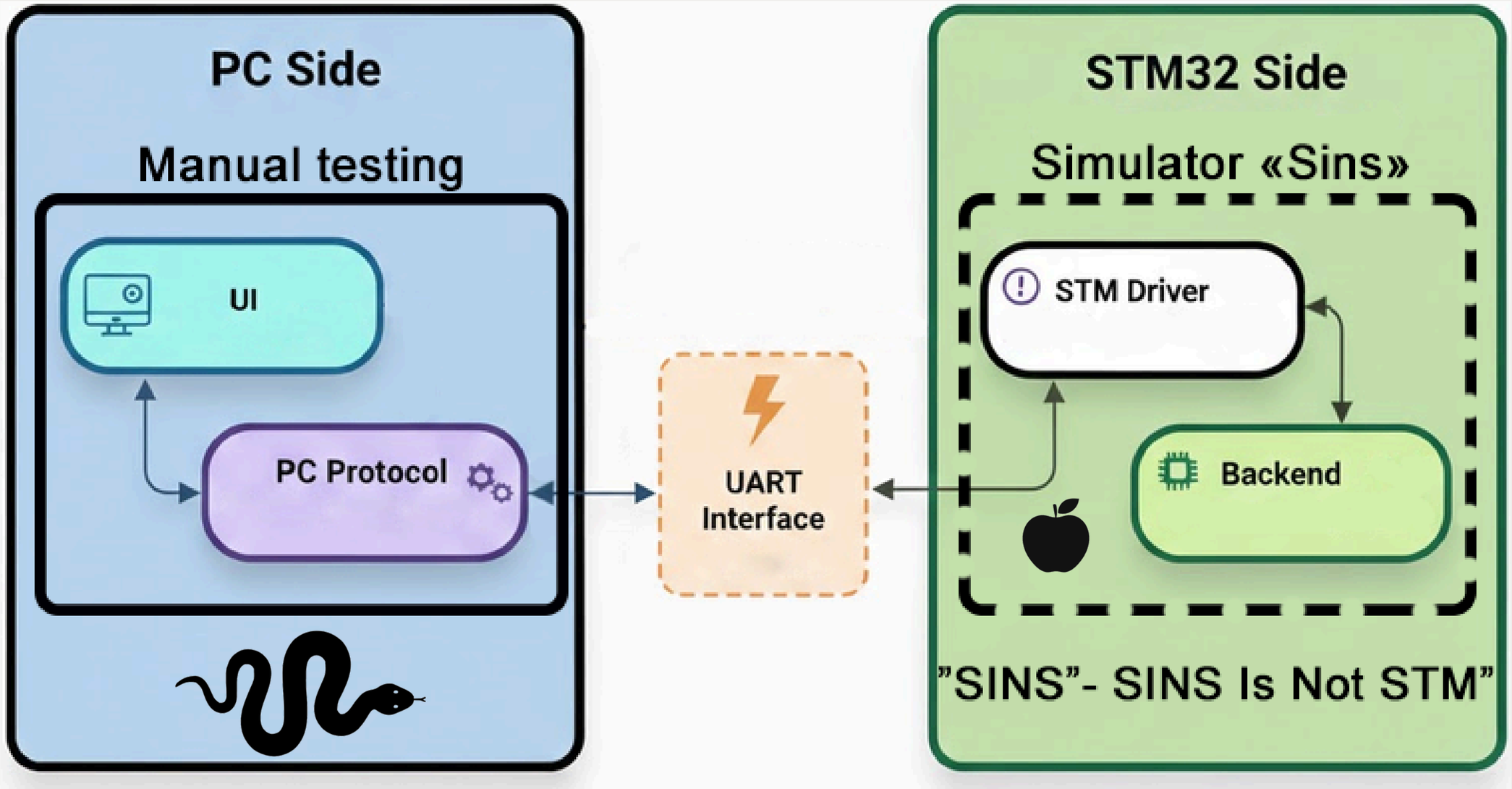
Show empty values

Filter by keyword or by field

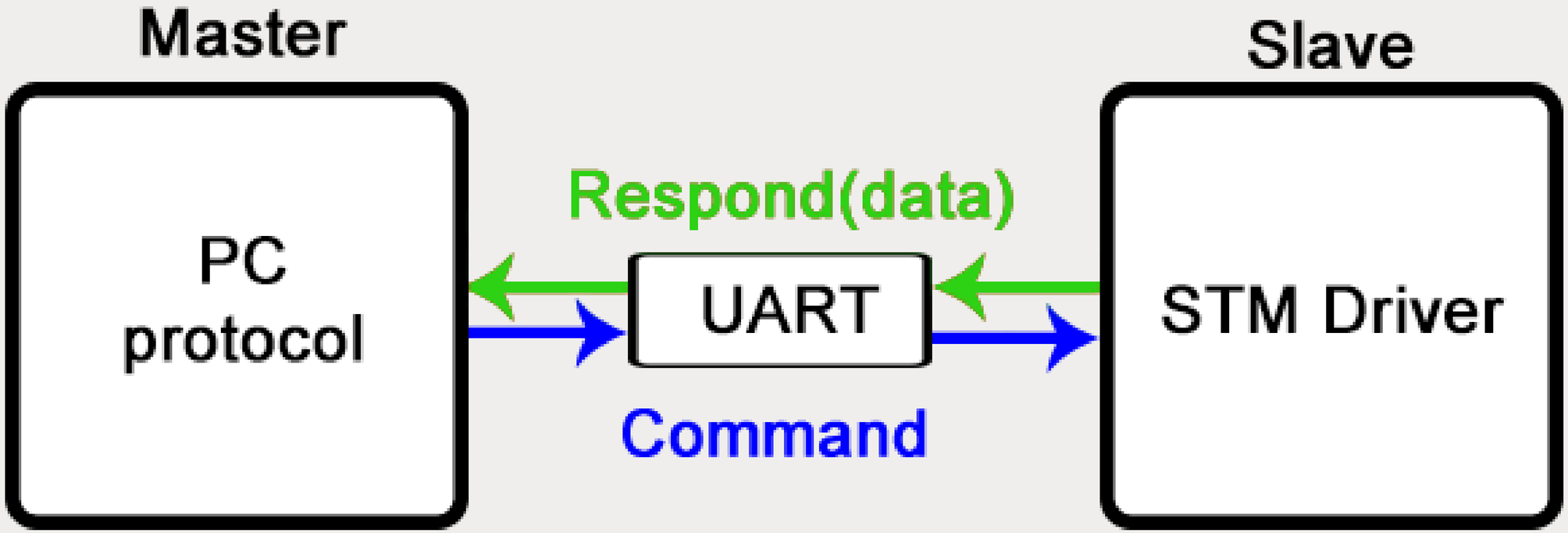
Done 31 Estimate: 0 This has been completed

	Title	Status
1	Supervise branch and MRs name convetion #18	Done
2	Create exe file #51	Done
3	Prepare demo presentation #52	Done
4	Design and implement error handling. #25	Done
5	Add configuration to for scanning COM port #28	Done
6	Implement auto-detection com port script #46	Done
7	UART driver code refactoring (STM32) #55	Done
8	Review MR #27 Move settings to controller #50	Done
9	Create task performance calendar file #58	Done
10	Bug fix test tool (missaligment in console output) #53	Done
11	Finalize technical document #54	Done
12	Design and implement obstacle changing from stm side #20	Done

Game structure



PC protocol and STM Driver



Backend

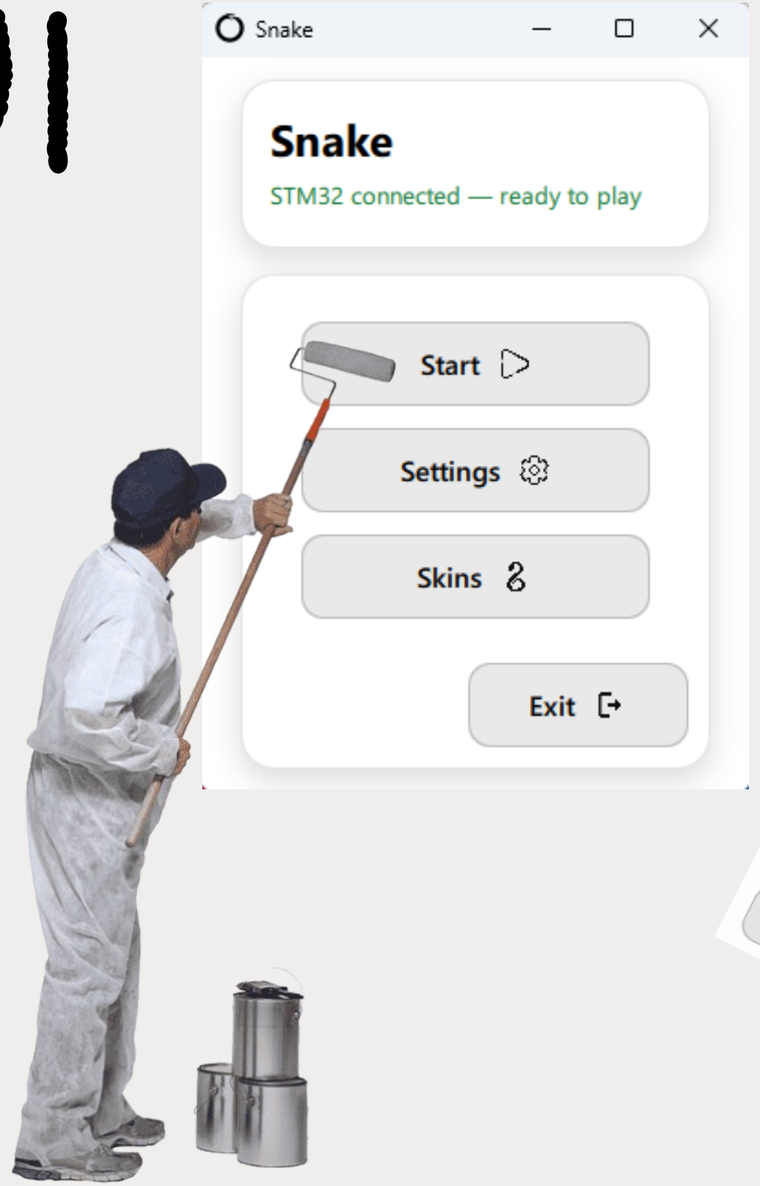
Core Logic
Snake movement
Collision detection
Level generation (0–5)
Score & growth

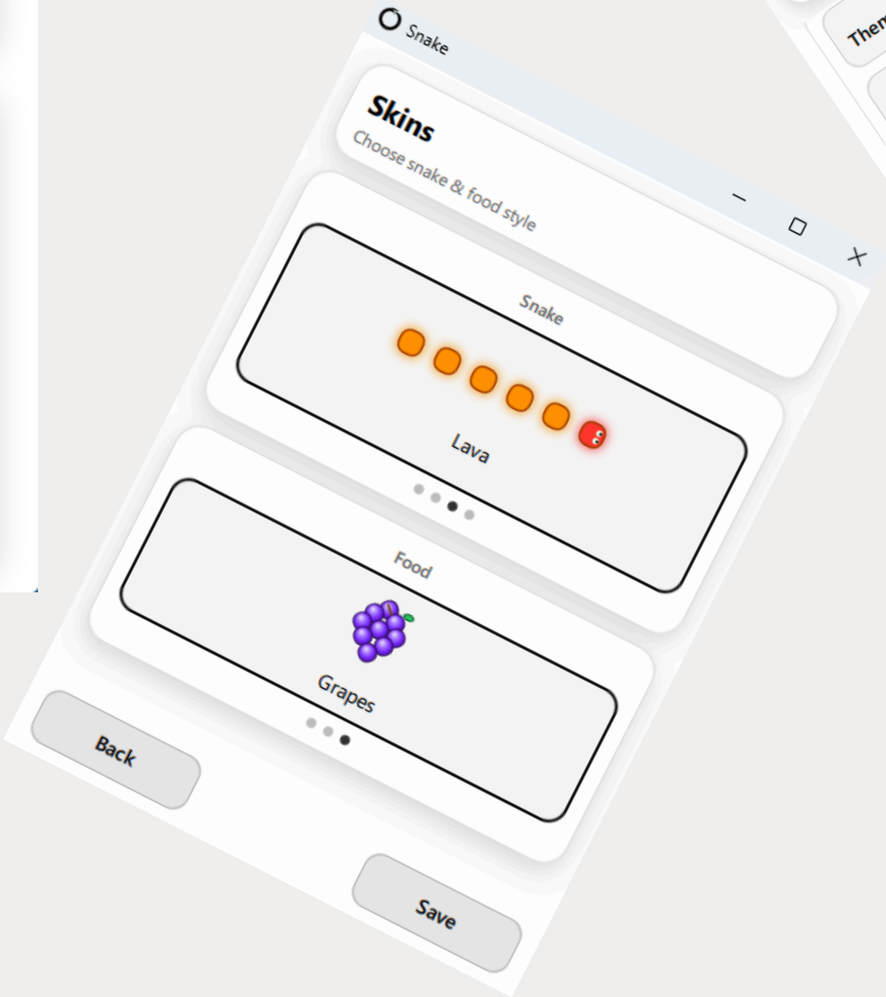
Game Loop
Tick: 100–400 ms
Input → Update → Send to PC
States: Menu / Playing / Pause / Game Over

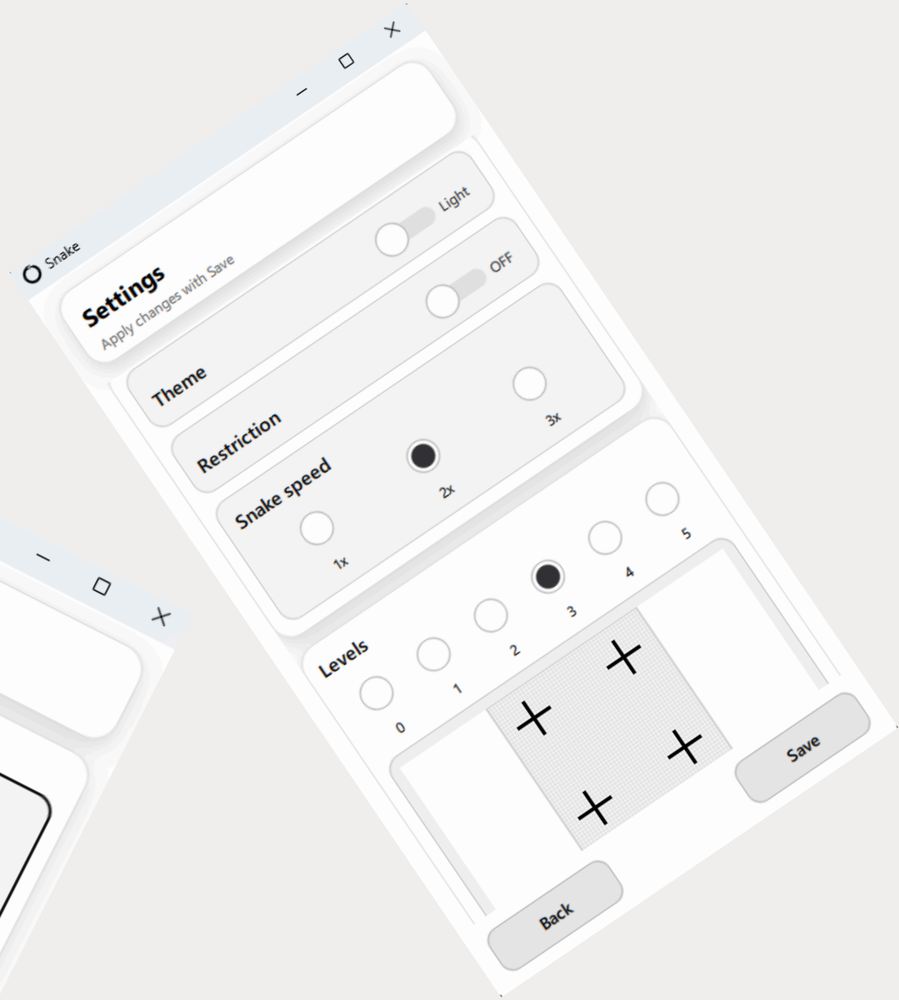
Features
Labyrinth levels
Speed control
Best score in Flash
Fixes
No 180° turns
Stable collisions

Literally me:

UI


Snake
Snake
STM32 connected — ready to play
Start
Settings
Skins
Exit


Snake
Skins
Choose snake & food style
Snake
Lava
Food
Grapes
Back
Save


Snake
Settings
Apply changes with Save
Light
OFF
Theme
Restriction
3x
Snake speed
2x
1x
5
4
3
2
1
0
Levels
Save
Back

Simulator “SINS”

```
G:○  MENU        B:○  |  dbg: OFF      |  btn: idle          Snake Simulator v3.2
CONSOLE                                 PC ──► STM32          STM32 ──► PC
┌─ Commands ──────◆
│ menu/play/pause/gameover/sleep
│ up/down/left/right
│ dbg 0..6
│ btn short/medium/long
│ btn press / btn release
│ speed slow/medium/fast
│ wrap / nowrap
│ send AA BB CC DD EE FF GG HH
│ cls  --  clear panels
└──────────
```

```
STM32 ──► PC
STATE PLAYING
SCORE 1
DELTA (25,25) food(7,4)
DELTA (26,25) food(25,25)
DELTA (27,25) food(26,25)
DELTA (28,25) food(27,25)
DELTA (29,25) food(28,25)
DELTA (30,25) food(29,25)
DELTA (31,25) food(30,25)
DELTA (32,25) food(31,25)
DELTA (33,25) food(32,25)
DELTA (34,25) food(33,25)
DELTA (35,25) food(34,25)
DELTA (36,25) food(35,25)
DELTA (37,25) food(36,25)
DELTA (38,25) food(37,25)
```

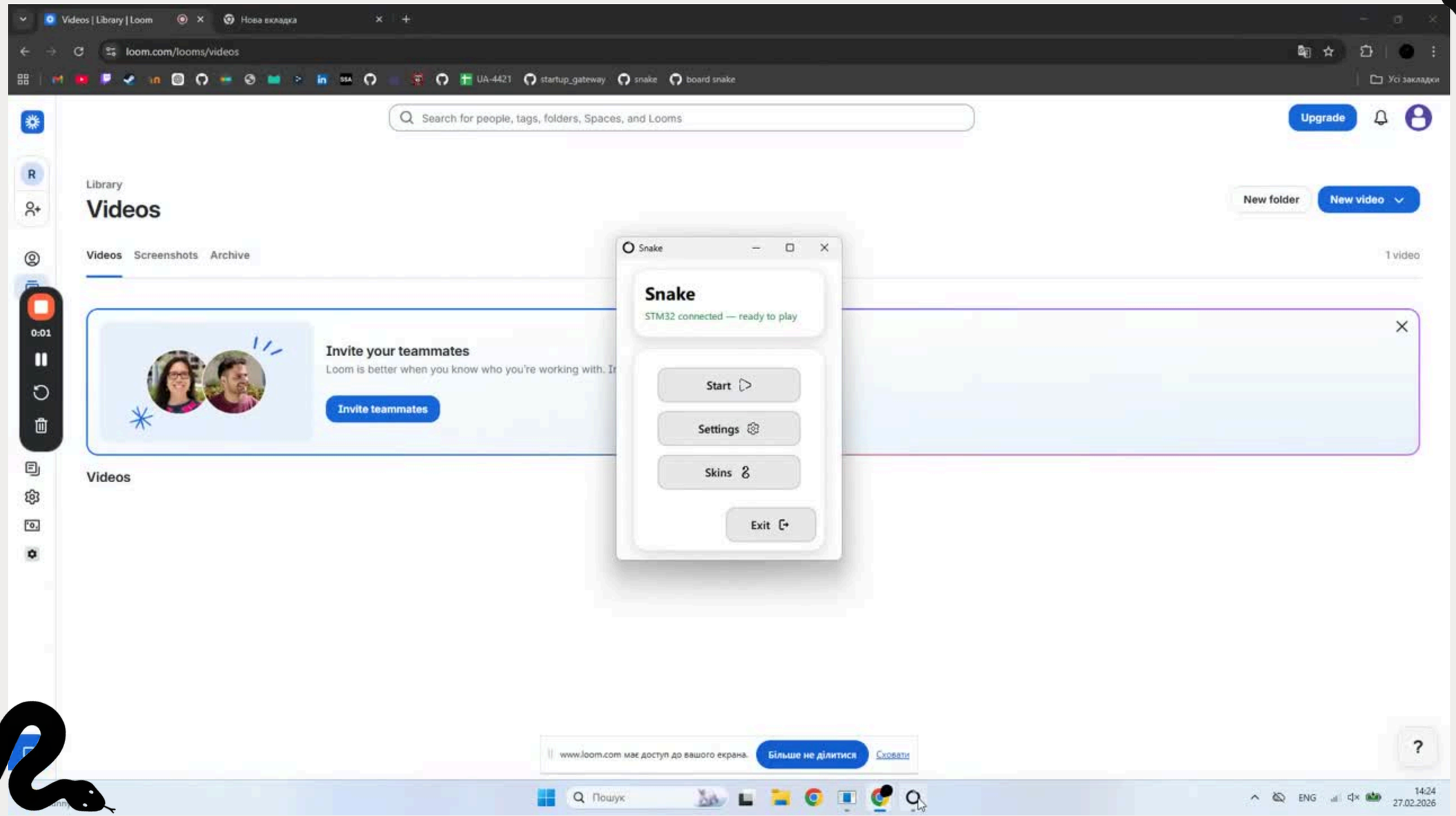
```
dbg: STEP      |  btn: idle
                 PC ──► STM32
                SET_STATE PLAYING
er/sleep        SET_STATE MENU
                MOVE LEFT
                MOVE RIGHT
                CONFIG [02=01]
e               DEBUG mode=6
```

```
G:●   GAMEOVER   B:○
```

Simulator “SINS”

```
C:\Users\Vasyan\STM32CubeIl

G:● PLAYING   B:○ | dbg: OFF   | level: 0| btn: idle   Snake Simulator v4.0

CONSOLE                          PC → STM32                STM32 → PC
 Commands ──────►                SET_STATE PLAYING         DELTA h(25,25) f/t(17,4)
 GAME CONTROL:                   MOVE LEFT                 SCORE len=1 val=0
   menu, play, pause, gameover,                            DELTA h(24,25) f/t(25,25)
   up, down, left, right                                   DELTA h(23,25) f/t(24,25)
                                                           DELTA h(22,25) f/t(23,25)
 LEVELS:                                                   DELTA h(21,25) f/t(22,25)
   level 0..5  or  lvl0..lvl5                              DELTA h(20,25) f/t(21,25)
   0=empty 1=corridors 2=rings                             DELTA h(19,25) f/t(20,25)
   3=crosses 4=maze 5=sym.corner                           DELTA h(18,25) f/t(19,25)

 WALL VISUALISATION:
   walls      - list all walls
   map        - show ASCII wall
   clearwalls - clear wall list

 DEBUG:
   dbg 0..6
   btn short/medium/long
   btn press / btn release

 CONFIG:
   speed slow/medium/fast
   wrap / nowrap

 TESTS:

►
```

Our game


Videos | Library | Loom
Нова вкладка
loom.com/looms/videos
Search for people, tags, folders, Spaces, and Looms
Upgrade
Library
Videos
New folder
New video
Videos
Screenshots
Archive
1 video
0:01
Invite your teammates
Loom is better when you know who you're working with.
Invite teammates
Snake
STM32 connected — ready to play
Start
Settings
Skins
Exit
Videos
www.loom.com має доступ до вашого екрана.
Більше не ділитися
Сховати
Пошук
ENG
14:24
27.02.2026

Conclusions

-Team work

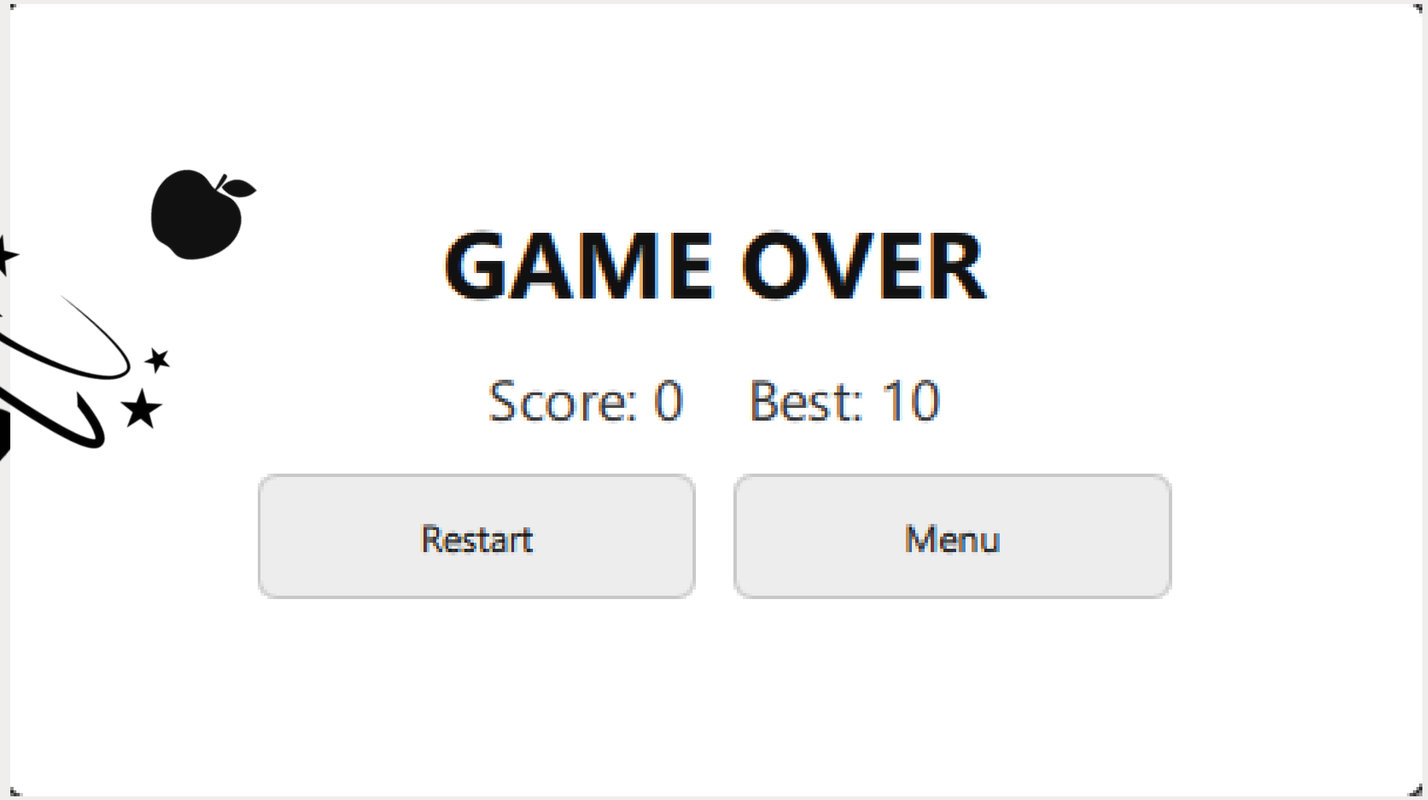
-"Commercial" project experience

-Organization of work

-Office work experience

-Improved hardware software skills

Thank you for your attention



Are there any questions?